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LERNAEAN KNOTS AND BAND SURGERY

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Abstract. The paper is devoted to a line of the knot theory related to the conjecture on the additivity of the crossing number for knots under connected sum. A series of weak versions of this conjecture are proved. Many of these versions are formulated in terms of the band surgery graph also called the $H(2)$-Gordian graph.

§1. Introduction

The present paper pertains to the classical theory of knots and develops a direction originating from the following well-known conjecture on the additivity of the crossing number for knots under connected sum.

Conjecture 1. *If a knot K is a connected sum of knots K_1 and K_2, then the crossing number* $\mathrm{cr}(K)$ *equals* $\mathrm{cr}(K_1) + \mathrm{cr}(K_2)$.

By now, this conjecture has been open for over a hundred years. A number of its significant weakenings are open as well. Various related conjectures, versions, and weakenings of Conjecture 1 form an extensive network with a complex structure of implications. In the present paper, we study a part of this network.

Let us formulate a chain of conjectures and definitions leading to our part of the network. The following conjecture is one of the most natural weakenings of Conjecture 1.

Conjecture 2. *The crossing number of a composite knot is not less than the crossing number of each of its summands.*

Conjecture 2 can be regarded as a conjecture about the dependence between two natural partial orders on the set of knots (links). These orders are "to be a summand" and "to have a smaller crossings number".

Definition (chimeras and cochimeras)**.** If a knot K is a summand in a composite knot N with $\mathrm{cr}(N) < \mathrm{cr}(K)$, we say that K is a *chimera* and N is a *cochimera accompanying* K. A link containing a cochimera accompanying K as a component is also called a *cochimera accompanying* K.

Conjecture 2 is (obviously) equivalent to the following one.

Conjecture $2'$**.** *The set of chimeras and the set of cochimeras are both empty.*

An option for the further weakening of Conjecture 2 relies on the following concept of λ-*regular* knots.

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Definition (λ-regular knots)**.** For a real number $\lambda > 0$, a knot K is said to be *λ-regular* if for each knot N containing K as a summand we have

$$\mathrm{cr}(N) \geq \lambda \cdot \mathrm{cr}(K).$$

In this terminology, chimeras are exactly non-1-regular knots. Therefore, Conjecture 2 is equivalent to the conjecture that all knots are 1-regular. The notion of λ-regularity gives a continual chain of "nested" (i. e., increasingly weaker) conjectures, for smaller and smaller λ, that all knots are λ-regular. For certain reasons, the following conjecture is of particular interest.

Conjecture 3. *All knots are $\frac{2}{3}$-regular.*

As far as the authors know, by now it has only been established that all knots are $\frac{1}{152}$-regular (this follows directly from the results of [52]).

Our part of the network of Conjecture 1 weakenings includes, in particular, conjectures that knots of certain classes are λ-regular. By now, 1-regularity has been proved for alternating knots, torus knots, and some other specific knot classes (for more details, see [61] and references therein, including [29, 23]).

Techniques developed in [12, 58, 75, 76, 47, 74] and properties of the Kauffman polynomial proved there can be used to show that the so-called *adequate* knots are 1-regular and *semi-adequate* knots are $\frac{1}{2}$-regular.[1] However, to our best knowledge, the existing literature lacks these conclusions. In this paper, we fill the gap and provide the corresponding proofs.

Theorem 1. *Every adequate knot is* 1*-regular.*

Theorem 2. *Every semi-adequate knot is $\frac{1}{2}$-regular.*

Lernaean knots. Another central object of the present paper is the class of Lernaean knots and their subspecies (this class is a subclass in the class of chimeras).[2] In the following, we rely on the standard concept of *crossing resolution* in a knot or link diagram. The crossing resolution is defined as a local transformation of the form  or  .

Definition (knot simplification)**.** A knot or link L is called a *simplification* of a knot K if L is obtained from K by a resolution of a crossing in a minimal diagram of K.

Definition (Lernaean knots)**.** A nontrivial knot K is said to be *Lernaean* if at least one of its simplifications contains K as a summand.

Definition (totally Lernaean knots)**.** A nontrivial knot K is said to be *totally Lernaean* if each simplification of K contains K as a summand.

Lernaean and totally Lernaean knots were introduced in [61] in the analysis of the conjecture on hyperbolic knot genericity.[3] In particular, in [61] it was proved that the assumption that the hyperbolic knots are generic implies that the set of totally Lernaean knots is nonempty. The properties of totally Lernaean knots seem so absurd that the

[1] Adequate and semi-adequate knots were introduced in [58, 76] as generalizations of alternating knots; these classes are relatively large; see §2 for definitions.

[2] We chose the term "Lernaean" because each Lernaean knot becomes more "complicated" in a certain sense when one tries to "simplify" it by resolving (cutting) a crossing in a diagram; cf. the properties of the Lernaean Hydra in Greek mythology (see, for example, "Hercules", Euripides): "I killed the hydra, that beast with the many heads that kept growing back again the moment I cut them off".

In fact, within the framework of Greek mythology, Lernaean Hydra and Chimera are sisters, but this does not introduce confusion in our case.

[3] This conjecture says that the proportion of hyperbolic knots among all of the prime knots of n or fewer crossings approaches 1 as n approaches infinity.

proof of the conjecture that their set is empty was viewed as the most promising way to refute the above-mentioned conjecture on hyperbolic knot genericity. In the course of further studies of Lernaean knots and their variations, the conjecture on hyperbolic knot genericity was indeed refuted (see [9]), but the question of the existence of totally Lernaean and Lernaean knots remains open. Continuing to develop this line of thought, we introduce neolernaean knots in this paper.

Definition (quasi-simplifications of links)**.** A link M is called a *quasi-simplification* of a link L if M is obtained from L by a resolution of a crossing in a (not necessarily minimal) diagram of L.[4]

Definition (neolernaean knots)**.** A nontrivial knot K is said to be *neolernaean* if each quasi-simplification L of K possesses at least one of the following properties (i. e., L is not simpler than K at least in one of the following senses):

(i) $\mathrm{cr}(L) \geq \mathrm{cr}(K)$,
(ii) L contains K as a summand.

The class of neolernaean knots lies in the set of totally Lernaean knots and continues the chain

$$\text{CHIMERAS} \supseteq \text{LERNAEAN} \supseteq \text{TOTALLY LERNAEAN} \supseteq \text{NEOLERNAEAN}.$$

These inclusions are obvious. Indeed,

- a knot is a chimera if and only if it has an accompanying cochimera,
- a knot is Lernaean if and only if it has an accompanying cochimera among its simplifications,
- a knot is totally Lernaean if and only if all of its simplifications are accompanying cochimeras,
- a knot is neolernaean only if its accompanying cochimeras contain a certain class containing all of its simplifications.

Note that the class of neolernaean knots is no longer associated with the conjecture on hyperbolic knot genericity. The defining property of neolernaean knots seems even more absurd than those of chimeras and Lernaean knots. However, the authors failed to find any well-known argument or construction that would directly imply that the class of neolernaean knots is empty. The described sequence of nested subclasses of chimeras continues, for example, by further strengthening the requirements on the size of the set of accompanying cochimeras, via generalizing the crossings resolution transformation, and/or by moving from single to multiple transformations. Based on the last of the mentioned possibilities, we introduce the following concept.

Definition (neolernaean knots levels)**.** Let K be a knot, and let $d \in \mathbb{N}$. A knot or link L is called a *d-fold quasi-simplification* of K if L is obtained by resolving d crossings in a (not necessarily minimal) diagram of K.[5] We say that a neolernaean knot K is a neolernaean knot *of level d* if each of its d-fold quasi-simplifications L has at least one of properties (i) and (ii), and there is a $(d+1)$-fold quasi-simplification of K having neither of these properties.

Obviously, for an arbitrary knot K, the unknot is a $(\mathrm{cr}(K)-2)$-fold quasi-simplification of K. It follows that the level of an arbitrary neolernaean knot K is at most $\mathrm{cr}(K) - 2$.

[4]It is easy to see that the binary relation "A is a quasi-simplification of B" is symmetric and reflexive.

[5]Adding a loop (type I Reidemeister move) to an appropriate diagram of K shows that each d-fold quasi-simplification of K is also a $(d+1)$-fold quasi-simplification of K.

The following theorem shows a certain relationship between the levels of neolernaean knots and their λ-regularity.

Theorem 3. (1) *No* 1*-regular knot is neolernaean.*
(2) *No* $\frac{2}{3}$*-regular knot is neolernaean of level at least* 3*.*
(3) *No* $\frac{1}{2}$*-regular knot is neolernaean of level at least* 6*.*

Corollary 1. *No semi-adequate knot is neolernaean of level at least* 6*.*

The first statement of Theorem 3 is a reformulation of the observation discussed above: all neolernaean knots are Lernaean, and all Lernaean knots are chimeras. Our proofs of the second and third statements of Theorem 3 are based on a new combinatorial-geometric technique and are not presented in this paper. We expect that further development of this technique will make it possible to extrapolate the statements of Theorem 3 to the cases of λ-regular knots with $\lambda < 1/2$ and to deduce a proof of the following conjecture by using the established $\frac{1}{152}$-regularity of an arbitrary knot.

Conjecture 4. *The set of neolernaean knots of level at least* 1000 *is empty.*

We introduce another subspecies of Lernaean knots. The question of whether this subspecies is empty served as a starting point for this paper.

Definition (ultralernaean knots)**.** A nontrivial knot K is said to be *ultralernaean* if each quasi-simplification of K contains K as a summand.

We regard the following theorem as a pivotal result of this paper.

Theorem 4. *The set of ultralernaean knots is empty.*

Theorem 4 follows directly from Theorems 5–10 below. Each of these theorems is a refinement of Theorem 4.

Theorem 5. *Any nontrivial knot K has a diagram D with a crossing X such that neither of the two ways to resolve X produces a diagram of a link containing K as a summand or coinciding with K.*

Band surgeries. Before discussing neo- and ultralernaean knots in more detail, we introduce the following concept.

Definition (band surgery)**.** Let L be a link in $\mathbb{R}^3$, let I denote the segment $[0, 1]$, and let $b\colon I \times I \to \mathbb{R}^3$ be an embedding such that

$$b(I \times I) \cap L = b(I \times \partial I).$$

Then we say that the link

$$M = (L \setminus b(I \times \partial I)) \cup b(\partial I \times I)$$

is obtained from L by the *band surgery* along the *band* $b(I \times I)$. (See Figure 1.)

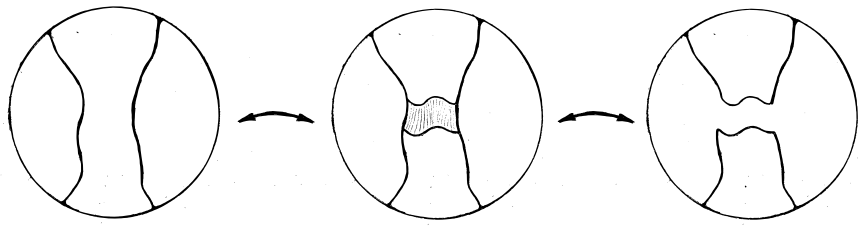


FIGURE 1

Definition (The $H(2)$-Gordian graph and $H(2)$-Gordian distance)**.** The *$H(2)$-Gordian graph* is the simple graph whose vertex set is the set of ambient isotopy classes of tame nonoriented links in $\mathbb{R}^3$ and two vertices are adjacent if the corresponding links are related by a band surgery. The number of edges in a shortest path between two vertices in the $H(2)$-Gordian graph defines a distance on the set of links. This distance is called the *$H(2)$-Gordian distance*.

If r is a nonnegative integer and L is a link, we denote by $B_r(L)$ the ball of radius r centered at L, that is, a link M is in $B_r(L)$ if and only if M and L are related by a chain of at most r band surgeries. We denote by $S_r(L)$ the sphere $B_r(L) \setminus B_{r-1}(L)$.

Among the array of fundamental works that laid the foundation for the systematic study of band surgery for knots and links, we rely primarily on the papers [56] and [32]. (In [32], band surgery was treated as a special case, for $n = 2$, of $H(n)$-moves.) Many authors have researched various properties of band surgery and related constructions over the past decade. See [40, 42, 43, 45] for more details on the relationship between band surgery and polynomial invariants, [13, 39, 41, 46, 77] for the $H(2)$-Gordian distance, [1, 2, 6, 7, 44, 62] for calculation of the $H(2)$-Gordian distance to the unknot, [59, 63] for the relationship between band surgery and lens spaces, [14] for fibered links band surgery, [33] for a description of cosmetic surgery, [35, 79, 80] for details on graphs and complexes associated with band surgery, and [34, 64] for applications in biology.

Definitions imply that a link Y is obtained from a link X by a resolution of a crossing in a diagram of X if and only if Y and X are related by a band surgery. This yields the following reformulations:

- the set of d-fold quasi-simplifications of an arbitrary knot K coincides with the ball $B_d(K)$,
- a nontrivial knot K is neolernaean if and only if each link L adjacent to K (i. e., each link in the sphere $S_1(K)$) either contains K as a summand or has $\mathrm{cr}(L) \geq \mathrm{cr}(K)$,
- a nontrivial knot K is ultralernaean if and only if each link in the sphere $S_1(K)$ contains K as a summand.

Thus, problems related to neolernaean and ultralernaean knots reduce to the study of neighborhoods of vertices in the $H(2)$-Gordian graph. The difficulty of describing objects of this kind is well known (see, for example, the results of [67, 68, 78] that each knot with unknotting number one is prime, which is equivalent to the theorem on the composite knots absence in the unit neighborhood of the unknot in the unknotting Gordian graph[6]).

Theorems 6–10 below describe properties of the spheres $S_1(K)$. The definitions of standard concepts of the knot and link theory used in these theorems are easy to find in the literature if necessary (see, for example, [66, 3, 57, 15]). Definitions of the least known concepts involved are given it the text.

Remark 1. It is easy to see that for any knot K, the sphere $S_1(K)$ contains connected sums of K with each knot (or link) lying in $S_1(U)$, where U stands for the unknot. The sphere $S_1(U)$ contains all two-strand windings of all knots.[7] Since we are primarily interested in the question of the existence of neolernaean and ultralernaean knots, further we focus on those elements of $S_1(K)$ that do not contain K as a summand.

[6]In this graph, the vertices corresponding to links L and M are adjacent if and only if L has a diagram such that some crossing change in this diagram gives a diagram of M.

[7]Two-strand windings are two-component links of genus 0 and knots of crosscap number 1, i. e., nontrivial knots that bound the Möbius strip embedded in $\mathbb{R}^3$; also, see the definitions of (p, q)-windings and of the crosscap number below.

Definition (crosscap number)**.** The *crosscap number* of a nontrivial knot K in $\mathbb{R}^3$ is the minimal value of the crosscap genus over all embedded compact connected nonorientable surfaces $F \subset \mathbb{R}^3$ with boundary $\partial F = K$ (in other words, this is the smallest number of Möbius bands needed to assemble such a surface). By definition, the crosscap number of the unknot is assumed to be zero.

Theorem 6. *If K is a nontrivial knot, then the sphere $S_1(K)$ contains*:

(1a) *a knot whose crosscap number is less than that of K*;
(1b) *a knot whose bridge number does not exceed that of K*;
(1c) *a knot whose arc index does not exceed that of K*;
(1d) *a knot whose braid index does not exceed that of K*;
(2a) *a two-component link whose genus is less than that of K*;
(2b) *a two-component link whose bridge number does not exceed that of K*;
(2c) *a two-component link whose arc index does not exceed that of K*;
(2d) *a two-component link whose braid index does not exceed that of K.*

None of these (possibly coinciding) knots and links contain K as a summand.

Item (1b) of Theorem 6 implies the following strengthening of Theorem 4, due to well-known properties of the bridge number [70, 71].

Corollary 2. *If K is a nontrivial knot, then $S_1(K)$ contains a knot that is not a satellite of K.*

Theorem 7. *Let K, K_1, and K_2 be knots, and let k be a nonnegative integer. Then the set $S_1(K)$ contains a two-component link L whose components are K_1 and K_2, and the linking number of the components is k. (If none of K_1 and K_2 coincides with K and none of K_1 and K_2 contains K as a summand, then L does not contain K as a summand.)*

Definition (knot windings)**.** A nontrivial knot Z is called a *winding* of a knot K if Z is ambient isotopic to a knot on the boundary of a tubular neighborhood of K, but not ambient isotopic to K.

Definition (slopes of curves)**.** Let V be a solid torus embedded in $\mathbb{R}^3$. There are two distinguished isotopy classes of simple noncontractible closed curves on the boundary torus ∂V. These are the meridional and longitudinal classes. We fix a meridian m and a longitude ℓ on ∂V and orient them in an arbitrary way. Then to any simple noncontractible (on ∂V) closed oriented curve γ on ∂V we assign its *slope*, which is the element p/q of the set $\mathbb{Q} \cup \{\infty\}$, where the integers p and q are determined by the following equation involving the homology classes $[\gamma]$, $[m]$, and $[\ell]$ of γ, m, and ℓ, respectively,

$$[\gamma] = p[\ell] + q[m].$$

Note that the slope does not depend on the orientation of γ, while the slope changes sign (keeping the absolute value) whenever the orientation of either longitude or meridian reverses. This means that the slope's absolute value is defined for nonoriented curves on ∂V as well.

Definition ((p, q)-windings)**.** A winding Z of a knot K is called a (p, q)-*winding* if Z is ambient isotopic to a knot γ on the boundary of a tubular neighborhood of K such that the absolute value of the slope of γ equals p/q for coprime positive integers p and q.[8] In some cases, the (q, p)-windings of nontrivial knots are referred to as *p-strand* windings.

[8]Note that a (p, q)-winding of the unknot (i. e., a torus knot of type (p, q)) is also a (q, p)-winding of the unknot.

Theorem 8. *For an arbitrary knot K, the set $S_1(K)$ contains all of the windings of K with the slope's absolute value belonging to the set*

$$\left\{ \frac{3+2i}{|2+(3+2i)j|} \;\middle|\; i,j \in \mathbb{Z}; i \geq 0 \right\}.$$

In particular, $S_1(K)$ contains all of the three-strand windings of K.

Definition (Brunnian link)**.** A nontrivial link is said to be *Brunnian* if all of its proper sublinks are trivial.

Theorem 9. *If K is a nontrivial knot, then the set $S_1(K)$ contains*:

(1) *a prime knot P that is a three-strand winding of K with*

$$\operatorname{cr}(P) \leq 9\operatorname{cr}(K) + 2;$$

(2) *a Brunnian (and therefore prime) two-component link L with*

$$\operatorname{cr}(L) \leq 4\sqrt[4]{6}^{\operatorname{cr}(K)} + 2.$$

For a link L, we denote by $\mathrm{C}_1(L)$ the number of knots (counting multiplicities) in the prime decomposition of L.

Theorem 10. *If K is a prime knot, then the set $S_1(K)$ contains a prime two-component link L with $\operatorname{cr}(L) \leq \operatorname{cr}(K) + 1$. More generally, if K is a nontrivial knot, then $S_1(K)$ contains a nonsplit two-component link L with $\operatorname{cr}(L) \leq \operatorname{cr}(K) + 1$ and $\mathrm{C}_1(L) < \mathrm{C}_1(K)$.*

Remark 2. In a certain sense, Theorems 5–10 outline a possible shape of the class of Lernaean knots by proving the emptiness for an extensive set of their subspecies. These theorems do not prove that the class of neolernaean knots is empty. However, each of these theorems (moreover, each of their items) is a strengthening or refinement of Theorem 4 on the ultralernaean class emptiness, because each of these items guarantees, for a nontrivial knot K, that $S_1(K)$ contains knots or links with various specific properties indicating that these knots and links do not contain K as a summand. Theorems 9 and 10 refine Theorem 4 in terms of the crossing number and show how close we can get to the condition $\operatorname{cr}(L) < \operatorname{cr}(K)$ of the definition of neolernaean knots if we are in the set of links that belong to $S_1(K)$ and do not contain K as a summand. To be more specific, Theorem 10 means that the set of quasi-simplifications of any nontrivial knot K contains a two-component link of at most $\operatorname{cr}(K)+1$ crossings and not containing K as a summand, and for establishing the neolernaean class emptiness it suffices to prove that the set of quasi-simplifications of an arbitrary nontrivial knot K contains a link of at most $\operatorname{cr}(K) - 1$ crossings and not containing K as a summand.

Remark 3. Theorem 6 involves only those knot invariants that detect the unknot and have controlled the behavior with respect to connected sum (the genus, crosscap number, bridge number, arc index, and braid index). Thus, each of the items of Theorem 6 implies Theorem 4.

Remark 4. Theorem 8 admits a generalization implying that for an arbitrary knot K, the set $S_1(K)$ contains a rich family of satellites of K that are not necessarily windings of K. These satellites are related to K by single band surgeries along bands lying in a tubular neighborhood of K.

Remark 5. An extension of an idea involved in the proof of Theorem 8 gives examples of windings of a given knot K that are contained in the balls $B_r(K)$ with $r > 1$.

Remark 6. An obstruction to extending Theorem 10 to the case of knots is the possibility that the knot obtained by the method described in the proof of the theorem coincides with a given knot.

Remark 7. All those knots and links in $S_1(K)$ that arise in Theorems 5–10 and do not contain the given knot K as a summand, are either certainly satellites of K (Theorem 8) or their class within Thurston's classification is not clear. This brings us to the following question.

Question 1. Show that for any nontrivial knot K, the sphere $S_1(K)$ contains hyperbolic knots.

Additional corollaries about the $H(2)$-Gordian graph. Theorems 6–10 and their proofs imply a number of properties of subsets in the $H(2)$-Gordian graph. These properties are less related to (neo)Lernaean knots and the crossing number additivity conjecture, but they are of some independent interest. We cite two such properties. To formulate them, we need several definitions.

Definition (band-surgery-connected subsets)**.** A subset in the set of knots and links is said to be *band-surgery-connected* if the corresponding induced subgraph[9] in the $H(2)$-Gordian graph is connected.

Theorem 11. *The following sets are band-surgery-connected*:

(1) *the set of all knots,*
(2) *the set of nontrivial knots,*
(3) *any cofinite subset in the set of all knots,*
(4) *the set of composite knots,*
(5) *the set of alternating knots,*
(6) *the set of torus knots* (*including the unknot*)*,*
(7) *the set of iterated torus knots* (*including the unknot*)*,*
(8) *the set of all knots that contain an arbitrary given knot K as a summand* (*including K*)*.*

Definition (weakly monotone functions)**.** A real-valued function f on the set of knots (links) is said to be *weakly monotone* (*with respect to band surgery*) if for each $C \in \mathbb{R}$, the preimage $f^{-1}(-\infty, C]$ is band-surgery-connected.

Obviously, the crossing number is weakly monotone on both the set of knots and the set of all links. At the same time, verification of the monotonicity for, say, the number of summands in the prime decomposition of a knot or link seems to be a difficult problem.

Conjecture 5. *The sets of prime and hyperbolic knots are band-surgery-connected.*

In parallel to Theorem 6, we prove the following.

Theorem 12. (1) *The crosscap number is weakly monotone on the set of knots.*
(2) *The bridge number and arc index are weakly monotone both on the set of knots and on the set of links.*
(3) *The braid index and genus are weakly monotone on the set of links.*

Structure of the paper. The paper is organized as follows.

In §2, we recall the notions of adequate and semi-adequate knots, introduce the concept of virtual height for links and prove Theorems 1 and 2.

In §3, we prove Theorems 5, 6, and 12.

In §4, we prove Theorem 7.

In §5, we prove Theorems 8 and 11.

In §6, we study meander and semi-meander knot diagrams and prove Theorem 9.

In §7, we introduce and study the notion of decompositionally essential crossings in knot diagrams and prove Theorem 10.

[9] A subgraph Γ' of a graph Γ is called *induced* if Γ' is formed by a subset of vertices of Γ and all those edges of Γ that connect pairs of vertices of this subset.

§2. ($\frac{1}{2}$-)Regularity of (semi-)adequate knots: proofs of Theorems 1 and 2

(Semi-)adequate links. For crossings in link diagrams, resolutions of the forms  and  are called *A-resolutions* and *B-resolutions*, respectively. A diagram D of a link L is said to be *A-adequate* if the number of circles in the diagram obtained from D by applying A-resolution at all of its crossings is greater than the number of circles in each of the diagrams obtained from D by B-resolution at one of its crossings and A-resolution at all remaining crossings. Switching the symbols "A" and "B" in this definition yields the definition of *B-adequate* diagrams. A diagram that is both A-adequate and B-adequate is called *adequate*. A link is said to be *adequate* if it admits an adequate diagram. A link is said to be *semi-adequate* if it admits either an A-adequate or B-adequate diagram.

The notion of adequate links was introduced in [58]. It generalizes the notion of alternating links. Most importantly, adequate diagrams are minimal (see [76]). The properties of (semi-)adequate links were studied, in particular, in [5, 26, 27, 36, 37, 38, 48, 53, 54, 55, 65, 72, 73, 74]. The class of semi-adequate links is relatively wide and includes, *inter alia*, all positive and negative links, the Alexander closures of all positive and negative braids and of all braids of index 3 (see [73]), all semi-alternating links and all Montesinos links (see [58, 73, 74]), and all knots with 10 or fewer crossings[10] (see [74, Inadequate examples], cf. [76, p. 290]), as well as flat windings of the members of all specified classes (see [58]); see, for example, the section titled "Which links are semi-adequate?" in [26] and the literature therein for details.

The Kauffman polynomial. We derive Theorems 1 and 2 (on the regularity of adequate and the $\frac{1}{2}$-regularity of semi-adequate knots) from known properties of the *Kauffman* and the *Brandt–Lickorish–Millett–Ho polynomial*, where the latter is a simplification of the former. The properties we need were proved in [12, 58, 75, 76, 47, 74]. The Kauffman polynomial $F_L(a, z)$ is a two-variable Laurent polynomial. It is an ambient isotopy invariant of oriented links (orientation is irrelevant in the case of knots). Many papers define the Kauffman polynomial in terms of a certain polynomial $\Lambda_D(a, z)$, which is an invariant of *regular isotopies* of nonoriented diagrams. This invariant polynomial $\Lambda_D(a, z)$ also takes values in $\mathbb{Z}[a^{\pm}, z^{\pm}]$ and is uniquely determined by the following set of properties (see [47, Definition 2.2] for more details).

- If D and D' are regularly isotopic diagrams (in the sense that D and D' are related by a sequence of type II and type III Reidemeister moves), then $\Lambda_D(a, z) = \Lambda_{D'}(a, z)$;
- For each quad of diagrams that are identical everywhere except at small fragments whose forms are shown in brackets below, we have[11]

$$\Lambda[\text{⨯}] + \Lambda[\text{⨯}] = z\Lambda[)(] + z\Lambda[\asymp];$$

- For each pair of diagrams, we have

$$\Lambda[\text{⨯}\bigcirc] = a\Lambda[)\,],$$
$$\Lambda[\text{⨯}\bigcirc] = a^{-1}\Lambda[)\,];$$

- The polynomial $\Lambda_{\bigcirc}(a, z)$ equals 1.

If an oriented diagram D represents a link L, then the value of the Kauffman polynomial $F_L(a, z)$ is

$$F_L(a, z) := a^{-w(D)}\Lambda_D(a, z),$$

where $w(D)$ is the *writhe number* of D.

[10] All knots with seven or fewer crossings are alternating.

[11] In the following formulas, $\Lambda[x]$ stands for the polynomial $\Lambda_D(a, z)$ of a diagram D with a fragment x.

Here, there is no need to recall a definition of the writhe number or other details of the above constructions because we derive Theorems 1 and 2 directly from the properties of polynomials proved in [47, 12, 75, 76, 74].

Virtual height of links. In order to structure the proofs of Theorems 1 and 2, we introduce a new auxiliary link invariant. This invariant is one of many possible invariants arising as characteristics of configurations, on the index lattice $\mathbb{Z} \times \mathbb{Z}$, of nonzero coefficients of the Kauffman polynomial.

Definition (virtual height)**.** Let $P = \sum_{r,s\in\mathbb{Z}} u_{rs} a^r z^s$ be a nontrivial Laurent polynomial. We define its *virtual maximal z-degree* $\hbar_z$ as

$$\hbar_z(P) := \frac{\max\{r+s \mid u_{rs} \neq 0\} + \max\{-r+s \mid u_{rs} \neq 0\}}{2}.$$

By the *virtual height* of an (oriented) link L we mean $\hbar_z(F_L(a,z))$. Since the Kauffman polynomial of any link is nonzero[12], it follows that the virtual height is well defined for any (oriented) link L. The formula implies that multiplying a polynomial by a does not change its virtual maximal z-degree. Thus, the virtual height is invariant under the change of any of the link components orientation. The *virtual height of a nonoriented link* is the virtual height of corresponding oriented links.

Remark 8. While this is not relevant to the following proof, we remark that the virtual height of any link is an integer. This follows, e. g., from [75, Proposition 3, assertion (ii)] stating that the coefficient u_{rs} of the polynomial $\Lambda_D(a,z) = \sum_{r,s\in\mathbb{Z}} u_{rs} a^r z^s$ of any diagram D with n crossings is nonzero only if the sum $n + r + s$ is even.

Remark 9. The virtual height of an (oriented) link L coincides with the virtual height of its mirror image L' because $F_{L'}(a,z) = F_L(a^{-1},z)$.

Next, we prove several properties of the virtual height. We need these properties for the proofs of Theorems 1 and 2.

Proposition 1. (1) *The virtual height of an arbitrary link is nonnegative.*
(2) *The virtual height is additive under connected sum.*
(3) *The virtual height of an arbitrary link L is less than or equal to* $\operatorname{cr}(L)$.
(4) *The virtual height of an adequate link L equals* $\operatorname{cr}(L)$.
(5) *The virtual height of a semi-adequate link L is at least* $\frac{1}{2}\operatorname{cr}(L)$.

Proof of Proposition 1. (1) The polynomial $F_L(1,z)$ equals the Brandt–Lickorish–Millett–Ho polynomial $Q_L(z)$ (see, for example, [47, p. 421]). For an arbitrary link L, the polynomial $Q_L(z) - 1$ is divisible by $2(z-1)$ (see [12, Property 2]), so the coefficient of z^0 in $Q_L(z)$ is odd (see [12, Corollary (ii)]). Hence, for some $r \in \mathbb{Z}$, the coefficient of $a^r z^0$ in $F_L(a,z)$ is nonzero. By the definition of the virtual height, this implies the required statement.

(2) The Kauffman polynomial is multiplicative under connected sum (see, for example, Lemma 2.8 in [47]) and is nonzero (as explained above). Since the ring of multivariable Laurent polynomials with integral coefficients is an integral domain, it follows that the *northwest edge* of the Kauffman polynomial is well defined and also multiplicative under connected sum. The northwest edge of the Kauffman polynomial is defined for a link with the Kauffman polynomial $\sum_{r,s\in\mathbb{Z}} u_{rs} a^r z^s$ as the polynomial

$$\sum_{\{(r',s')\in\mathbb{Z}\times\mathbb{Z} \mid r'+s' = \max\{r+s \mid u_{rs}\neq 0\}\}} u_{r's'} a^{r'} z^{s'}.$$

[12]In order to see this, it suffices to observe that for any link L we have $F_L(1,1) = 1$.

This means in particular that the integer-valued invariant of oriented links defined by using the Kauffman polynomial $F_L(a,z) = \sum_{r,s\in\mathbb{Z}} u_{rs}a^r z^s$ via the formula $\max\{r+s \mid u_{rs} \neq 0\}$ is well defined and additive under connected sum. By a similar argument, the invariant defined by the formula $\max\{-r+s \mid u_{rs} \neq 0\}$ is well defined and additive under connected sum. This also implies the desired additivity of the virtual height, because it is defined as the half-sum of the indicated additive invariants.

(3) If a diagram D of a link L has n crossings, then the polynomial $\Lambda_D(a,z) = u_{rs}a^r z^s$ has the following property (see [75, Theorem 4]):

$$|r| + s \leq n \text{ for all } r, s \text{ with } u_{rs} \neq 0.$$

Since $F_L(a,z) = a^{-w(D)}\Lambda_D(a,z)$, this proves the statement.

(4) If a diagram D of a link L has n crossings and is A-adequate, then for $\Lambda_D(a,z) = u_{rs}a^r z^s$ there are r and s such that $u_{rs} \neq 0$ and $r+s = n$, while if D is B-adequate, then there are r and s such that $u_{rs} \neq 0$ and $-r+s = n$ (see [76]). Since $F_L(a,z) = a^{-w(D)}\Lambda_D(a,z)$, the statement follows.

(5) If a diagram D of a link L has n crossings and is semi-adequate, then for $F_L(a,z) = u_{rs}a^r z^s$ there are $r, s \in \mathbb{Z}$ such that $u_{rs} \neq 0$ and $s \geq n/2$ (this is a consequence of [74, Proposition 3.1]). Then the inequality $n \geq \mathrm{cr}(L)$ implies the required statement. □

Proof of Theorems 1 *and* 2. Let L be a link of virtual height $\hbar(L)$, and let K be a knot of virtual height $\hbar(K)$. Assume that L contains K as a summand. If K is adequate, then by Proposition 1 the following inequalities hold (the numbers above the equality and inequality symbols are the numbers of related statements in Proposition 1):

$$\mathrm{cr}(L) \overset{(3)}{\geq} \hbar(L) \overset{(1),(2)}{\geq} \hbar(K) \overset{(4)}{=} \mathrm{cr}(K).$$

If K is semi-adequate, then the following inequalities hold:

$$\mathrm{cr}(L) \overset{(3)}{\geq} \hbar(L) \overset{(1),(2)}{\geq} \hbar(K) \overset{(5)}{\geq} \frac{1}{2}\,\mathrm{cr}(K).$$

This proves the regularity of adequate knots and the $\frac{1}{2}$-regularity of semi-adequate knots. □

§3. Proofs of Theorems 5, 6, and 12

Proofs of Theorem 5 *and of items* (1b) *and* (2b) *of Theorem* 6. Let K be a nontrivial knot with bridge number $b(K)$, and let $\mathrm{cb}(K)$ be the minimal number of crossings over all diagrams having precisely $b(K)$ bridges and representing K[13]. Let D be a diagram representing K and having precisely $b(K)$ bridges and precisely $\mathrm{cb}(K)$ crossings. Let B be a bridge in D, and let x be a crossing in D such that B ends at x with an undercrossing. The minimality assumptions imply that no bridge forms a loop in D, so that the bridge having an overcrossing at x is distinct from B. Therefore, if K' and L are the knot and the two-component link, respectively, that are obtained from D by resolutions at x, then neither K' nor L has bridge number greater than $b(K)$. Moreover, the assumption that D has precisely $\mathrm{cb}(K)$ crossings guarantees that $K' \neq K$. Thus, both K' and L belong to $S_1(K)$, which implies items (1b) and (2b) of Theorem 6. Since a connected sum of a knot and a link that is not the unknot increases the bridge number ([70, 71]), this implies Theorem 5 as well. □

[13]Recall that a *bridge* on a link diagram is an arc that includes at least one overcrossing, has no undercrossings, and is not contained in a larger arc with these properties. In addition, all simple closed curves with no crossings are bridges.

Remark 10. In a similar way, Theorem 5 follows from the proof of items (1c) and (2c) of Theorem 6 given below.

Proof of item (1a) *of Theorem* 6. Let K be a nontrivial knot in $\mathbb{R}^3$, and let k be its crosscap number. (Since K is nontrivial, it follows that $k \geq 1$.) By the definition of the crosscap number, $\mathbb{R}^3$ contains an embedded compact connected nonorientable surface F of crosscap genus k with boundary $\partial F = K$. For every positive integer k, a connected compact nonorientable surface of genus k with nonempty connected boundary is unique up to homeomorphism and is obtained by gluing k disorienting bands to the boundary of the disk. Let us represent F in this form (a disk and bands) and consider a band surgery of K along one of these bands. The resulting knot K' is the boundary of an embedded surface that is the union of a disk with $k-1$ disorienting bands, so that the crosscap number of K' is at most $k-1$. It remains to observe that K' belongs to $S_1(K)$. □

Proof of item (2a) *of Theorem* 6. Let K be a nontrivial knot, in $\mathbb{R}^3$, of genus g. Then according to the definition of the knot genus, $\mathbb{R}^3$ contains an embedded connected compact orientable surface F of genus g with boundary $\partial F = K$. Since K is nontrivial, it follows that g is positive, so that F contains a band B such that the surgery along B does not split F and transforms K to a two-component link L bounding a surface of genus $g-1$. Thus, L lies in $S_1(K)$, and its genus is at most $g-1$. □

Proof of items (1c) *and* (2c) *of Theorem* 6. We say that two arcs on two pages of a book representation of a knot or link are *decussating* if the endpoints of these arcs are located on the binding axis in the order $1-2-1-2$, and not $1-1-2-2$ or $1-2-2-1$. Let K be a nontrivial knot. Among all those book representations of K that have the smallest number of pages, we choose a representation $\mathcal{B}$ that has the smallest number of decussating pairs of arcs.[14] Note that only the unknot has a book representation without decussating arcs (because an ambient isotopy moves all the arcs of such a representation into one half-plane). We also note that in the case where two arcs on two adjacent pages of a book representation do not decussate, they can be "swapped" (by transferring one of these arcs – by an ambient isotopy – to the page of another one and *vice versa*, without moving the rest of the knot). It follows that $\mathcal{B}$ contains decussating arcs and that we may, without loss of generality, assume that the arcs of some pair of decussating arcs (say, A and B) are located on adjacent pages. We consider the "superposition" of these adjacent pages and a resolution of the resulting crossing at the arcs A and B. We see that there is a pair of band surgeries each of which transforms the arcs A and B into two nondecussating arcs (say, C and D, where we mean that our two band surgeries have distinct "C and D") located on the same two pages and with the same quad of endpoints as for A and B, so that the resulting book representation for one of these two band surgeries corresponds to a knot, say K', while the resulting book representation for another one corresponds to a two-component link, say L. Neither the arc index of K' nor that of L exceeds the arc index of K. Moreover, $K' \neq K$ because the number of decussating pairs of arcs in the resulting book representation for K' is at least one less than that of $\mathcal{B}$. (Indeed, any arc decussating with both C and D decussate with both A and B, and any arc decussating with precisely one of C and D decussate with precisely one of A and B.) This argument completes the proof. □

Proof of item (1d) *of Theorem* 6. For each $n \geq 2$, we denote by ex the homomorphism $B_n \to \mathbb{Z}$ defined by the condition $\mathrm{ex}(\sigma_i) = 1$, where σ_i is a standard Artin generator (see,

[14]The minimal number of decussating pairs is needed for item (1c), in the case of two-component links – item (2c) – the number of decussating pairs can be disregarded.

for example, [3, 18], [60, §1.3]). The number $\mathrm{ex}(\beta)$, for $\beta \in B_n$, is called the *exponential sum* or the *algebraic number of crossings* of β (see [24]).

Figure 2 shows that for any $n \geq 2$ and any $\beta \in B_n$, the Alexander closure [4] of β is related to the Alexander closure of the braid $\beta\sigma_1^4$ by a band surgery.

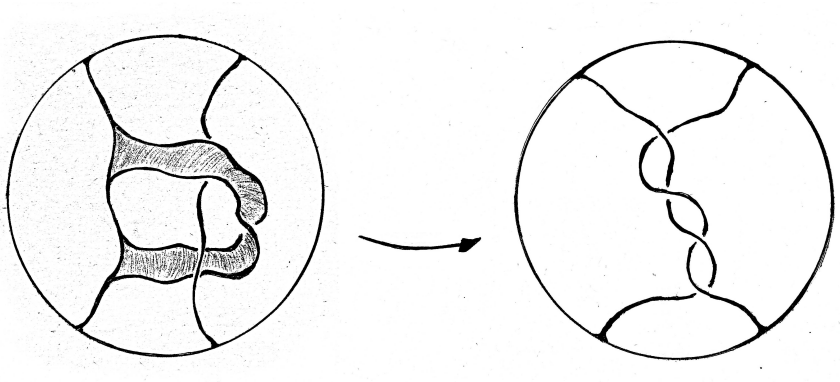


FIGURE 2

For a nontrivial knot K, choose a braid, β, of the smallest braid index among the braids whose Alexander closure is K. Then the knot K' represented by the braid $\beta\sigma_1^4$ does not coincide with K because, by virtue of the Jones conjecture proved in [24], the exponential sum value for the minimum index braid representing a given knot is an invariant of unoriented knots (considered up to ambient isotopy, but not up to mirror symmetry). See [24]. Thus, K' lies in $S_1(K)$ and the braid index of K' does not exceed that of K. □

Proof of item (2d) *of Theorem* 6. To verify this assertion, it suffices to resolve (in an appropriate way) any crossing in the projection of any braid of minimum index representing a given knot. □

Proof of the final statement of Theorem 6. The crosscap number does not decrease under the connected sum with any knot, and the genus does not decrease under the connected sum with any link. The bridge number, the arc index, and the braid index increase under the connected sum with any link that is not the unknot. The listed properties are proved, for example, in [16, 57, 70, 71, 17, 11]. The statement follows. □

Proof of item (1) *of Theorem* 12. Item (1a) of Theorem 6 implies that if K is a knot of crosscap number k then the $H(2)$-Gordian graph contains a path P between K and the unknot such that the length of P is at most k, and each vertex of P is a knot having crosscap number at most k. This immediately implies the required statement. □

Proof of item (2) *of Theorem* 12. We introduce a *secondary* type link invariant cb defined as the minimal number of crossings over all diagrams with the smallest number of bridges representing a given link. Arguments in the proof of items (1b) and (2b) of Theorem 6 show that, in the $H(2)$-Gordian graph, any nontrivial link L is adjacent either to a link with a smaller bridge number or to a link with the same bridge number, but with a smaller value of the invariant cb. This implies by induction that there is a path P from L to a trivial link such that no vertex of P is a link whose bridge number exceeds that of L. Transforming a trivial link into the unknot by surgeries, we find that for an arbitrary link M with bridge number b, the $H(2)$-Gordian graph contains a path P'

from M to the unknot such that no vertex of P' is a link whose bridge number exceeds b. This shows that the bridge number is weakly monotone on the set of links.

In the above construction, if we start with a diagram of a knot and each time when resolving a crossing we choose the way to resolve that produces a knot (not a two-component link), then we see that the bridge number is weakly monotone on the set of knots as well.

The proof for the arc index is similar to the above proof for the bridge number; in the case of the arc index, our secondary type invariant is the minimal number of decussating pairs of arcs over all book representations with the smallest number of pages (decussating arcs were defined in the proof of items (1c) and (2c) of Theorem 6). □

Proof of item (3) *of Theorem* 12. Let L be a link that is not the unknot, and let b be the braid index of L. Then there exists a diagram β of L having the form of a b-strand closed braid. By resolving one by one all crossings in β such that each of the involved diagrams still has the form of a closed braid, we get a path in the $H(2)$-Gordian graph from L to the trivial b-component link such that no vertex of this path is a link whose braid index exceeds b. Repeating the argument of the above proof for the bridge number completes the proof for the braid index.

Let L be a link (in $\mathbb{R}^3$) that is not the unknot, and let g be the genus of L. By the definition of the genus, $\mathbb{R}^3$ contains an embedded (connected compact orientable) surface F of genus g with boundary $\partial F = L$. It is easy to see that if $g > 0$, then F contains a band B such that the band surgery along B gives a link whose genus is at most $g-1$ and whose number of components is greater by one than that of L; and if L is not a knot, then F contains a band B such that the surgery along B gives a link whose genus is at most g and whose number of components is less by one than that of L. This implies that for an arbitrary link L of genus g, the $H(2)$-Gordian graph contains a path from L to the unknot such that no vertex of this path is a link whose genus is greater than g. This proves the weak monotonicity of the genus on the set of links. □

Remark 11. The methods used in the proof of Theorem 12 do not give a proof of the weak monotonicity of the braid index on the set of knots. (The authors do not know whether the braid index is weakly monotone on the set of knots.) In particular, the use of the exponential sum (in the proof of item (1d) of Theorem 6) as a secondary invariant – by analogy with the approach in the proof of item (2) of Theorem 12 – does not prove the weak monotonicity of the braid index on the set of knots, because minimizing the parameters associated with the exponential sum does not lead to a knot with a lower braid index.

§4. Proof of Theorem 7

We present a proof of Theorem 7 based on results obtained in [50]. These results are related to the theory of spatial graphs. Recall that (topological) *graphs* are one-dimensional CW complexes. Zero-dimensional cells of topological graphs are called *vertices*, and one-dimensional ones are called *edges*. The *theta-graph*, or *θ-graph*, is a topological graph with two vertices and three edges I_0, I_1, and I_2 none of which is a loop. In [50] it was proved that for any triple of knots K_0, K_1, K_2 there is an embedding of the θ-graph $f\colon \theta \to \mathbb{R}^3$ such that the images $f(I_1 \cup I_2)$, $f(I_0 \cup I_2)$, and $f(I_0 \cup I_1)$ are realizations of K_0, K_1, and K_2, respectively. This implies the main statement of Theorem 7. Indeed, if the images $f(I_1 \cup I_2)$, $f(I_0 \cup I_2)$, and $f(I_0 \cup I_1)$ are realizations of the knots $K_0 := K$, K_1, and K_2, respectively, and if B is an appropriately located and appropriately twisted "thin" band stretched along the arc $f(I_0)$, then the band surgery along B transforms $K_0 = f(I_1 \cup I_2)$ to a two-component link L whose components are K_1 and K_2 and whose

linking number is k. Theorems [69, 30, 31] on the uniqueness of the prime decomposition of knots and links imply the additional statement of the theorem due to the fact that if a link contains a knot K_0 as a summand, then at least one of the components of this link must also contain K_0 as a summand.

§5. Proof of Theorems 8 and 11

Proof of Theorem 8. Let K be a knot in $\mathbb{R}^3$, let V be a tubular neighborhood of K, and let T be a winding of K such that T lies in the boundary torus ∂V and the slope absolute value of T equals $(3+2i)/2$ for some $i \in \mathbb{Z}$, $i \geq 0$. We fix a meridian m and a longitude ℓ on ∂V such that m intersects T at exactly $3+2i$ points, ℓ intersects T at exactly two points, ℓ intersects m at exactly one point, and $\ell \cap m \cap T = \emptyset$. Then $\ell \cap T$ splits ℓ into two arcs. Let J be that one of these two arcs which does not meet m. Let $b\colon I \times I \to \partial V$ be an embedding of a square such that $b(\{1/2\} \times I) = J$ and

$$b(I \times \{0\}) \cup b(I \times \{1\}) = b(I \times I) \cap T.$$

Let T' denote the curve obtained from T by the band surgery along $b(I \times I)$. Then T' does not meet ℓ while the intersection $T' \cap m$ coincides with $T \cap m$ and consists of $3+2i$ points. Since $3+2i$ is odd, this means that T' is not contractible on ∂V and isotopic to ℓ, which, in turn, is isotopic to K. Thus, T lies in $S_1(K)$.

Now, let Z be a winding of K contained in ∂V such that the slope absolute value of Z is $\frac{3+2i}{|2+(3+2i)j|}$ for some $i, j \in \mathbb{Z}$, $i \geq 0$. Clearly, ∂V contains a winding S of K such that the slope absolute value of S is $(3+2i)/2$ and the $|j|$-fold Dehn twist of ∂V along m transfers S into a winding S_D isotopic to Z. We observe that this $|j|$-fold Dehn twist along m turns ℓ into a curve ℓ_D also isotopic to K. Without loss of generality, we may assume that $b(I \times I)$ does not intersect a neighborhood of m supporting the twist. This means that the band surgery along $b(I \times I)$ is well defined for S_D. It is easy to see that this surgery turns S_D into a curve isotopic to ℓ_D, which is isotopic to K. Therefore, S_D belongs to $S_1(K)$, whence it follows that Z belongs to $S_1(K)$ because S_D is isotopic to Z. Thus, we have shown that $S_1(K)$ contains all windings of K with slope absolute value belonging to the set $\left\{ \frac{3+2i}{|2+(3+2i)j|} \;\middle|\; i, j \in \mathbb{Z}; i \geq 0 \right\}$. The theorem is proved. □

Proof of item (1) *of Theorem* 11. The band-surgery-connectedness of the set of knots immediately follows, in particular, from the obvious fact that for any nontrivial knot K, the set $S_1(K)$ contains a knot whose crossing number is less than that of K. □

Proof of item (8) *of Theorem* 11. Let Z be a knot in $\mathbb{R}^3$. Suppose that Z is a connected sum of two knots Q and K_0. This means that $\mathbb{R}^3$ contains a closed ball B such that the pair $(B, B \cap Z)$ is a one-strand tangle knotted into Q. Then taking the quotient of $\mathbb{R}^3$ by B turns Z into K_0. Since the $H(2)$-Gordian graph is connected, it follows that there is a set of bands such that band surgeries along these bands turn K_0 into the unknot. We can choose these bands in such a way that none of them contains the point corresponding to B. This implies that there is a set of bands for Z such that none of these bands intersect B and band surgeries along these bands turn Z into Q. In addition, Schubert's theorem [69] on the uniqueness of the prime decomposition of knots implies that each knot corresponding to a vertex of the path from Z to Q either is ambient isotopic to Q or contains Q as a summand. This implies the required result. □

Proof of item (3) *of Theorem* 11. Let X be an arbitrary finite set of knots. Note that the set $S_1(U)$, where U is the unknot, is infinite (for example, because $S_1(U)$ contains all double-strand windings of all nontrivial knots, see also [79]). Therefore, $S_1(U)$ contains a knot Q such that the set $[Q]$ consisting of Q and all knots containing Q as a summand

does not intersect X. By virtue of item (8) proved above, $[Q]$ is band-surgery-connected. In addition, the technique presented in the proof of item (8) shows that (since $Q \in S_1(U)$) an arbitrary knot K and a knot that is a connected sum of K and Q are related by a band surgery. Hence it follows that any set of knots containing $[Q]$ is band-surgery-connected. In particular, the complement $\mathscr{K} \setminus X$, where $\mathscr{K}$ is the set of all knots, is also band-surgery-connected. This is the desired conclusion. □

Proof of item (2) *of Theorem* 11. This follows directly from item (3) proved above. □

Proof of item (4) *of Theorem* 11. By virtue of item (8) proved above, for any knot K, the set $[K]$ (consisting of K and all knots containing K as a summand) is band-surgery-connected. This implies item (4) because for any two knots A and B (we are currently interested in the case of composite knots) the sets $[A]$ and $[B]$ do intersect. □

Proof of item (5) *of Theorem* 11. Clearly, resolving a crossing in an alternating diagram always produces an alternating diagram with fewer crossings. This easily implies the statement. □

Proof of item (6) *of Theorem* 11. Let V be an unknotted solid torus in $\mathbb{R}^3$, and let T be a knot contained in ∂V. For a simple closed curve C on ∂V, we define the *index* of C as the minimal number of intersection points of C with a meridian, taken over all meridians on ∂V. Let $p := p(T)$ be the index of T. If $p \le 1$, then T is the unknot. If $p \ge 2$, then we fix a meridian m on ∂V such that m intersects T at exactly p points. The intersection points of T and m divide m into p arcs. Let J be any of these arcs. Let $b\colon I \times I \to \partial V$ be an embedding such that $b(\{1/2\} \times I) = J$ and

$$b(I \times \{0\}) \cup b(I \times \{1\}) = b(I \times I) \cap T.$$

Let T_1 be the link obtained from T by the band surgery along $b(I \times I)$. Since we have assumed that the number of points in $m \cap T$ is minimal, it follows that T_1 is a knot (not a two-component link). Note that T_1 intersects m at exactly $p - 2$ points, so T_1 has a lower index than T.

The above arguments show that each nontrivial torus knot T of type (p, q), where $p \ge 2$, is related by a band surgery either to the unknot or to a torus knot of type (p_1, q_1) with $p_1 < p$. Induction on p completes the proof. □

Proof of item (7) *of Theorem* 11. Let us say that the unknot is the *iterated torus knot of order* 0. For $n \ge 1$, a knot K is called an *iterated torus knot of order* n if K is a winding of some iterated torus knot of order $n - 1$. Let T be an iterated torus knot of order $n \ge 1$. Then T is a winding of some iterated torus knot T_{n-1} of order $n - 1$. We may assume that T is located on the boundary of a tubular neighborhood V of T_{n-1}. Applying the argument of the proof of item (6) to the case of T and V, we see that T is related by a sequence of band surgeries along bands embedded in ∂V either to the unknot or to a knot isotopic to T_{n-1}. Induction on n completes the proof. □

§6. Proof of Theorem 9

Proof of item (1) *of Theorem* 9. It is easy to see that if we "triple" each of the arcs in a minimal diagram of a knot K, then we get a diagram with $3^2 \operatorname{cr}(K) = 9 \operatorname{cr}(K)$ crossings representing a three-component link such that each of the three components of this link is knotted into K (this construction is described in detail, for example, in [58, §2]). If we make an obvious transformation (which adds two crossings) of any of diagram sections consisting of three "parallel" arcs, then we get a diagram with $9 \operatorname{cr}(K) + 2$ crossings representing a knot that is a three-strand winding of K. It remains to observe that all knots that are three-strand windings of K belong to $S_1(K)$ by Theorem 8. □

Proof of item (2) *of Theorem* 9. To prove this item, we use a certain technique involving meander diagrams of knots. A knot diagram is said to be *semi-meander* if it is composed of two simple (i. e., without self-intersections) smooth arcs. In addition, if the common endpoints of these arcs both lie on the boundary of the unbounded component of the complement of the diagram, then this diagram is said to be *meander*.

The proof is based on the fact that each knot has a meander diagram. The mentioned fact of existence has been independently proved by many authors by using various techniques (see [10] for a detailed historical background on this issue). We move on to the proof of item (2).

Let K be a nontrivial knot, and let D be a meander diagram of K. By definition, D is composed of two simple smooth arcs whose common endpoints (we denote these endpoints by a and b) belong to the boundary of the unbounded component of the complement of D (see the example in Figure 3 (a)). Take a topological disk B in the plane containing D such that the interior of B intersects D along two simple arcs containing a and b (see Figure 3 (b)). Then do a type II Reidemeister move on these arcs with support inside B. We can resolve any of the two new crossings on the resulting diagram in a suitable way and obtain a two-component Brunnian link (see Figure 3 (d)). (In Figure 3, we present one of the two possibilities where the simple arcs forming D remain simple after performing the Reidemeister move. It is easy to see that the statement also holds true if each of the arcs acquires one self-intersection after this transformation.)

Since it is known that K has a semi-meander diagram with at most $\sqrt[4]{6}^{\mathrm{cr}(K)}$ crossings (see [8]), the following lemma completes the proof. □

Lemma 1. *If a knot has a semi-meander diagram with n crossings, then the same knot has a meander diagram with at most $4n$ crossings.*

Proof of Lemma 1. Let D be a semi-meander diagram with n crossings composed of two simple arcs γ_0 and γ_1 whose common endpoints (none of which is a crossing of the diagram) are denoted by a and b. Without loss of generality we may assume that a lies on the boundary of the unbounded component of the complement of D, that γ_0 is a segment, and that each arc of γ_1 that is free of crossings intersects the line L containing the segment γ_0 at one point at most. Let $L_1 \subset L$ be the ray with endpoint at a containing γ_0. We choose a point c on L_1 so that the intersection of D with L_1 is a subset of the segment with endpoints a and c. Let δ_0 denote the segment with endpoints b and c. Clearly, the relative interior of δ_0 intersects γ_1 at n points at most. Now, choose a point d on γ_1 such that the arc $\beta \subset \gamma_1$ with endpoints b and d does not contain diagram crossings. It is easy to see that there exists a simple arc δ_1 with endpoints d and c such that δ_1 does not intersect γ_1 and no crossing-free subarc of γ_0 intersects δ_1 at more than one point. Furthermore, note that δ_1 can be selected so that it intersects δ_0 at no more than n points. Now, we construct a new diagram D' by replacing the arc β in D with the union of the curves δ_0 and δ_1 (with smoothings at the endpoints) and arranging overcrossings/undercrossings so that δ_0 has overcrossings at all of the crossings involved, and δ_1 has overcrossings at all crossings except for crossings with δ_0. As a result, we get a meander diagram with at most $4n$ crossings representing the same knot. □

§7. Proof of Theorem 10

The proof of Theorem 10 involves the following notion.

Definition (decompositionally essential crossings in knot diagrams)**.** Let D be a diagram of a nontrivial knot, and let x be a crossing in D. We place D on a plane $\mathbb{R}^2$ located in $\mathbb{R}^3$. Let Γ be a knot in $\mathbb{R}^3$ corresponding to D in the sense that D is a regular orthogonal

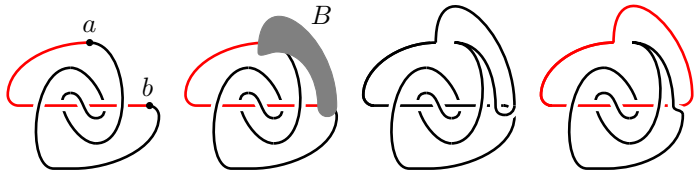


FIGURE 3

projection for Γ. Let I_x denote a *chord* of Γ corresponding to x in the sense that I_x projects to x, where a chord is a segment in $\mathbb{R}^3$ with endpoints on Γ. We say that a closed topological ball B in $\mathbb{R}^3$ is a *local knotting domain* for Γ if the pair $(B, B \cap \Gamma)$ is a nontrivial single-stranded tangle. Let $B_1, \dots, B_k$ be a finite set of pairwise disjoint balls each of which is a local knotting domain for Γ. We say that this set of balls is a (*geometric*) *connected sum decomposition scheme* for Γ if Γ is a connected sum of knots $K_1, \dots, K_k$, where K_i is the knot obtained by factorizing the tangle $(B_i, B_i \cap \Gamma)$ over the sphere ∂B_i. We say that a crossing x is *decompositionally inessential* if there exists a connected sum decomposition scheme for Γ such that none of the balls of the scheme intersect I_x. Otherwise, we say that x is *decompositionally essential*.

Remark 12. We do not require the knots in the local knotting domains be prime, so that, in particular, any nontrivial knot has a connected sum decomposition scheme formed by a single ball. Furthermore, if a crossing x (in the notation of the above definition) is decompositionally inessential, so that there are connected sum decomposition schemes for Γ that do not intersect I_x, then there exists such a scheme consisting of one or two balls, because by adding tubular neighborhoods of knot subarcs we can merge, into a single local knotting domain, all of the balls of a given scheme that intersect the same arc in the pair of arcs $\Gamma \setminus I_x$.

Remark 13. The papers [51, 19, 20, 22, 21] studied concepts close to the concept of decompositionally essential crossings. Compare Lemma 3 below and Theorem 2.31 in [22].

Conjecture 6. *The crossings of any minimal diagram of an arbitrary knot are decompositionally essential.*

Note that Conjecture 6 is a weakening of Conjecture 2.

Lemma 2. *If a knot is not a chimera, then all crossings of all minimal diagrams of this knot are decompositionally essential.*

Proof. Suppose that a minimal diagram D of a nontrivial knot K contains a decompositionally inessential crossing x. Let L denote the two-component link obtained from D by resolving x, and let K_1 and K_2 denote the components of L. Then $\mathrm{cr}(L) \le \mathrm{cr}(K) - 1$. We place D on a plane $\mathbb{R}^2 \subset \mathbb{R}^3$. Let Γ and I_x be the same as in the definition of decompositionally essential crossings. Since x is decompositionally inessential, there is a connected sum decomposition scheme $\mathcal{S}$ for Γ such that none of the balls of $\mathcal{S}$ intersect I_x. Without loss of generality we assume that $\mathcal{S}$ consists of one or two balls (by at most one ball for each of the two arcs of $\Gamma \setminus I_x$, see Remark 12). For $i = 1, 2$, we denote by Γ_i that one of the two arcs of the set $\Gamma \setminus I_x$ which projects into the component K_i in L, and by A_i we denote the knot corresponding to the local knotting that $\mathcal{S}$ cuts out on Γ_i. Then

(i) K_1 contains A_1 as a summand,
(ii) K_2 contains A_2 as a summand,
(iii) K is a connected sum of A_1 and A_2.

Therefore, the following inequalities hold:

$$\mathrm{cr}(K_1) + \mathrm{cr}(K_2) \le \mathrm{cr}(L) \le \mathrm{cr}(K) - 1 < \mathrm{cr}(K) \overset{\text{(iii)}}{\le} \mathrm{cr}(A_1) + \mathrm{cr}(A_2).$$

Consequently, at least one of the following inequalities hold:

$$\mathrm{cr}(K_1) < \mathrm{cr}(A_1) \quad \text{and} \quad \mathrm{cr}(K_2) < \mathrm{cr}(A_2).$$

This means that at least one of A_1 and A_2 is a chimera. In addition, (i)–(iii) imply that a suitable connected sum of K_1 and K_2 contains K as a summand. We denote this connected sum by N. Thus, since

$$\mathrm{cr}(N) \le \mathrm{cr}(K_1) + \mathrm{cr}(K_2) < \mathrm{cr}(K),$$

we see that K is also a chimera. □

We do not use Lemma 2 in our proof of Theorem 10 directly. This lemma is presented in order to make our constructions clearer. In order to prove Theorem 10, we need the following lemma (compare with [9, Lemma 1] and [22, Theorem 2.31]).

Lemma 3. *Each diagram of a nontrivial knot contains a decompositionally essential crossing.*

Proof of Lemma 3. Let K be a knot, and let D be a diagram of K. We place D on a plane $\mathbb{R}^2$ located in $\mathbb{R}^3$. Let Γ be a simple closed curve in $\mathbb{R}^3$ representing K such that D is the image of Γ under a projection to $\mathbb{R}^2$. We assume without loss of generality that Γ does not intersect $\mathbb{R}^2$. For any crossing x on D, we denote by e_x that one of the two points of Γ projecting to x which is closer to x. Let E be the union of e_x over all crossings x of D. The projections of connected components of the set $\Gamma \setminus E$ to $\mathbb{R}^2$ will be called the *overarcs* of D. We say that two overarcs α and β *meet* at a crossing x if x lies on α and x is a limit point for β.

Now, we choose an orientation of D and consider the corresponding Wirtinger presentation for the knot group, i. e., the fundamental group $\pi_1(\mathbb{R}^3 \setminus \Gamma)$ (a detailed description of Wirtinger presentations can be found, for example, in [15]). Let x be a crossing in D, let α and β be overarcs of D meeting at x, let a and b be the Wirtinger generators corresponding to α and β, respectively, and let I_x denote the segment in $\mathbb{R}^3$ with endpoints at Γ that projects to x. The construction of the Wirtinger presentation implies that any loop representing the element ab^{-1} is freely homotopic to a simple closed curve Δ contained in an arbitrarily small neighborhood of I_x. Note that any loop representing ab^{-1} (and, therefore, Δ too) has zero linking number with Γ. If x is decompositionally inessential, then by definition there is a connected sum decomposition scheme $B_1, \dots, B_k$ for Γ such that none of $B_1, \dots, B_k$ intersect I_x. Observe that in this case, the union $\Gamma \cup B_1 \cup \dots \cup B_k$ is ambient isotopic to the unknot with k "beads" (or "thickenings"), i. e., the complement

$$V := (\mathbb{R}^3 \cup \{+\infty\}) \setminus (\Gamma \cup B_1 \cup \dots \cup B_k)$$

is isotopic to an open solid torus. Since none of $B_1, \dots, B_k$ intersect I_x, we can choose Δ sufficiently close to I_x and assume that Δ lies in V. This implies that Δ is contractible (in V and, therefore, in $\mathbb{R}^3 \setminus \Gamma$). Hence the loops representing ab^{-1} are contractible, so that the element is trivial, that is, $a = b$. Thus, the Wirtinger generators corresponding to overarcs meeting at a decompositionally inessential crossing represent the same element of the knot group. Consequently, if all crossings of a knot diagram are decompositionally inessential, then all Wirtinger generators represent the same element of the knot group.

This is possible only if our knot group is cyclic. But the knot group is cyclic if and only if this knot is the unknot (see, e. g., the proof in [15, Proposition 3.17]). □

Remark 14. A slightly more careful analysis shows that each diagram of a nontrivial knot has at least three decompositionally essential crossings.

Proof of Theorem 10. Denote by D be a minimal diagram for a nontrivial knot K. By Lemma 3, D contains at least one decompositionally essential crossing (say, x). Up to rotations and reflections, we may assume that the resolution of x that turns D into a diagram of a two-component link has the form $\times \to \asymp$. Let L_0 and L_2 be the links whose diagrams are obtained by applying, to D at x, the transformations

$$\times \to \asymp \quad \text{and} \quad \times \to \rtimes,$$

respectively. Then L_2 is also two-component. Note that

$$\operatorname{cr}(L_0) \le \operatorname{cr}(K) - 1 \quad \text{and} \quad \operatorname{cr}(L_2) \le \operatorname{cr}(K) + 1.$$

Since $\times \to \rtimes$ and $\rtimes \to \times$ are mutually inverse transformations, and $\rtimes \to \times$ is a crossing resolution, it follows that both L_0 and L_2 lie in $S_1(K)$. Hence, to complete the proof, it suffices to show that at least one of L_0 and L_2 is nonsplit and has fewer one-component terms in its prime decomposition than K has.

To verify this, we construct realizations of K, L_0, and L_2 in $\mathbb{R}^3$ by placing the plane $\mathbb{R}^2$ that contains D in $\mathbb{R}^3$ and taking, in $\mathbb{R}^3$, a closed curve Γ whose orthogonal projection to $\mathbb{R}^2$ is D (the same construction as in the definition of decompositionally essential crossings). We may assume that Γ coincides with D everywhere except for small arcs corresponding to overcrossings and undercrossings at the crossings of D. Then the chord $I_x \subset \mathbb{R}^3$ that projects to x and whose endpoints lie on Γ is "relatively small" and lies in some Euclidean ball $X \subset \mathbb{R}^3$ centered at x. It is clear from the construction that we can select parameters in such a way that the pair $(X, X \cap \Gamma)$ is a trivial two-strand tangle whose arcs project to $\mathbb{R}^2$ in the form of $\times$. Thus, if we transform Γ by replacing its subset $X \cap \Gamma$, which is a pair of arcs, by pairs of arcs whose projections to $\mathbb{R}^2$ have the forms $\asymp$ and $\rtimes$ (with the same quad of endpoints in both cases), then we obtain realizations of L_0 and L_2, respectively.

A transformation of a link $L \subset \mathbb{R}^3$ is called a *rational tangle replacement* if this transformation replaces a trivial two-strand tangle $(B, B \cap L)$, with a ball $B \subset \mathbb{R}^3$, by another trivial two-strand tangle. Thus, each two of K, L_0, and L_2 are related by a rational tangle replacement. At the level of double branched covers, the rational tangle replacements correspond to Dehn surgeries: we remove a solid torus from a manifold and glue it back along some automorphism of the boundary torus. The gluing map sends solid torus meridians to an isotopy class of curves (on the torus), which characterizes the surgery. The minimal number of intersection points of the original solid torus meridian and curves isotopic to the image of this meridian (this image is the meridian of the attached solid torus) will be called the *characteristic* of the Dehn surgery (and also of the rational tangle replacement that induces this surgery). A direct description of the characteristic in terms of tangles is given, for example, in [25]. A simple standard computation shows that the rational tangle replacement corresponding to the diagrammatic transformation $\asymp \to \rtimes$ and providing the transition $L_0 \to L_2$ has characteristic 2 (see, for example, [25] or [28]).

We will use the last fact later, and now we turn to the tangle $(Y, Y \cap \Gamma)$, where $Y = \mathbb{R}^3 \cup \{\infty\} \setminus \operatorname{int}(X)$. The next step of our proof is to construct, in $\mathbb{R}^3 \setminus X$, a set of pairwise disjoint local knotting domains for Γ such that taking the quotient by these domains turns $(Y, Y \cap \Gamma)$ into a locally trivial tangle. (Recall that a tangle (B, t) is called *locally trivial* or a tangle *without local knots* if there is no ball $Z \subset B$ such that the pair

$(Z, Z \cap t)$ is a nontrivial single-stranded tangle; see, for example, [49]. Nonsplit locally trivial tangles are said to be *prime*. A split two-strand tangle without local knots is trivial.)

Note that if we have a finite (possibly empty) collection E of pairwise disjoint local knotting domains in $\mathbb{R}^3 \setminus X$ for Γ such that the quotient by the balls of E turns $(Y, Y \cap \Gamma)$ into a tangle having local knots, then a ball W containing one of these local knots can be chosen lying in $\mathbb{R}^3 \setminus X$ (in order to see this, take any appropriate ball containing a local knot, draw the point $\{\infty\}$ outside the ball by an isotopy, and pull the ball away from the sphere $\partial X = \partial Y$). In addition, we may assume that W contains no point representing contracted balls of E (because one can draw these points out of W by a suitable isotopy if need be). Consequently, we can enlarge such a collection E with an additional nontrivial local knotting domain. Since all of the knots corresponding to disjoint local knotting domains are (disjoint) summands of the knot K, it follows that the set of collections of pairwise disjoint nondegenerate local knotting domains in $\mathbb{R}^3 \setminus X$ for Γ contains a maximal collection E' that cannot be enlarged. Therefore, taking the quotient by the balls of E' turns $(Y, Y \cap \Gamma)$ into a locally trivial tangle $(Y', Y' \cap \Gamma')$.

Since we assume that x is decompositionally essential while no ball of E' intersects X and hence no ball of E' intersects $I_x \subset X$, it follows that E' is not a connected sum decomposition scheme for Γ. This means in particular that the total number of knots in the prime decompositions of local knots corresponding to E' is less than the number of knots in the prime decomposition of K.

Let K', Λ'_0, and Λ'_2 denote the links obtained from Γ, Λ_0, and Λ_2, respectively, when we take the quotient by the balls of E'. Then Λ'_0 and Λ'_2 (as well as Λ_0 and Λ_2) are related by the rational tangle replacement $(X, X \cap \Lambda_0) \to (X, X \cap \Lambda_2)$ of characteristic 2. Furthermore, the complementary tangle $(Y', Y' \cap \Lambda'_0) = (Y', Y' \cap \Lambda'_2) = (Y', Y' \cap \Gamma')$ is locally trivial.

We have two cases:

(p) the tangle $(Y', Y' \cap \Gamma')$ is prime,
(t) the tangle $(Y', Y' \cap \Gamma')$ is trivial.

In case (p), at least one of Λ'_0 and Λ'_2 is prime by virtue of Corollary 1 in [25]. If $(Y', Y' \cap \Gamma')$ is trivial, then both Λ'_0 and Λ'_2 belong to the class of so-called rational (two-bridge) links and, obviously, differ in the linking number between the components. Therefore, in this case at least one of Λ'_0 and Λ'_2 is nontrivial and hence prime being two-bridged. So, we arrive at the conclusion that at least one of Λ'_0 and Λ'_2 (say, Λ'_i) is nonsplit and prime. Therefore, when decomposing Λ_i into prime summands, we obtain the two-component link Λ'_i and, as one-component summands, the same set of knots (with possible repetitions) as when decomposing into prime summands the local knots corresponding to E'. Thus, the number of one-component summands in the prime decomposition of Λ_i is less than that of K. □

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